

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Open-label, Study of Subcutaneous Nivolumab + Relatlimab Fixed-dose Combination versus Intravenous Nivolumab + Relatlimab Fixed-dose Combination in Participants with Previously Untreated Metastatic or Unresectable Melanoma **Study Title:** A Study of Subcutaneous Nivolumab + Relatlimab Fixed-dose Combination (FDC) in Previously Untreated Metastatic or Unresectable Melanoma **Short Title/Acronym:** RELATIVITY-127 **Protocol Number:** CA224-127 **NCT Number:** NCT05625399 **Posting ID:** 31774509539330535 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab + Relatlimab Fixed-dose Combination (FDC) **Phase of Development:** Phase III **Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** Head of the Global Submission Management - Clinical Trials (mg-gsm-ct@bms.com)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate that the study drug exposure level of the nivolumab + relatlimab FDC subcutaneous (SC) formulation is not worse than (non-inferior to) nivolumab + relatlimab FDC intravenous (IV) administration in participants with previously untreated metastatic or unresectable melanoma. **Secondary Objectives:** To compare the SC and IV administration routes for safety and efficacy. Specific parameters include Objective Response Rate (ORR), Duration of Response (DOR), and Disease Control Rate (DCR) by Blinded Independent Central Review (BICR), as well as peak (C_{max}) and time-averaged steady-state (C_{avgss}) serum concentrations.

2.2 Background & Rationale To address the clinical need for more convenient drug administration methods. Nivolumab and relatlimab are a synergistic dual immunotherapy combining anti-PD-1 and anti-LAG-3 mechanisms. Subcutaneous injections can drastically reduce treatment times, minimize infusion-related reactions, and alleviate healthcare system burdens compared to standard intravenous infusions. This trial evaluates if the SC route offers comparable systemic absorption and clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Intravenous Nivolumab + Relatlimab FDC) **Blinding:** Open-label **Randomization:** Randomized **Status:** ACTIVE_NOT_RECRUITING

3.2 Planned Enrollment & Duration Target **N**: 579 participants
Number of Sites: Global, ~275 locations (United States + 18 other countries)
Estimated Study Start: 05-Mar-2023 **Estimated Primary Completion:** 17-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 12 years. Adolescents (≥ 12 to < 18 years of age) must weigh ≥ 40 kg at the time of signing informed consent (assent). 2. Histologically confirmed Stage III (unresectable) or Stage IV (metastatic) melanoma, per the American Joint Committee on Cancer (AJCC) staging system (8th edition). 3. Measurable disease by computed tomography (CT) or magnetic resonance imaging (MRI) per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1). 4. Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 1 (or Lansky Performance Score $\geq 80\%$ for adolescents).

4.2 Exclusion Criteria 1. Ocular melanoma. 2. History of myocarditis, regardless of etiology. 3. Condition requiring systemic treatment with either corticosteroids (>10 mg daily prednisone equivalent) or other immunosuppressive medications within 14 days of start of study treatment. 4. Active, known, or suspected autoimmune disease. 5. Concurrent malignancy requiring treatment or history of prior malignancy active within 2 years prior to randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Information: nivolumab (ATC Code: L01FF01; Trade Names: Opdivo, Opdivo) + Relatlimab Fixed-Dose Combination **EMA URLs:** [Opdivo EPAR](#), [Opdivo EPAR](#) **Dosage/Frequency:** Nivolumab + Relatlimab Fixed-Dose Combination administered on a scheduled cycle. **Route of Administration:** Subcutaneous (SC) vs. Intravenous (IV). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Inhaled or topical steroids, and adrenal replacement steroid doses >10 mg daily prednisone equivalent in the absence of active autoimmune disease. Hormone replacement for hypothyroidism. **Prohibited:** Systemic immunosuppressants and high-dose systemic corticosteroids (≥ 10 mg prednisone equivalent) within 14 days of treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent,

Tumor Imaging (CT/MRI), Labs | | **Baseline** | Day 0 |
Randomization, First Dose (SC or IV) | | **Interim** | Day 28 /
Scheduled cycles | Safety Labs, PK Blood Draws (Cmin, Cmax),
Efficacy Assessment | | **End of Study** | Variable | Final Tumor
Assessment, Exit Interview, Safety Follow-Up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Time-averaged serum concentration over 28 days after the first dose (Cavgd28) of Nivolumab 2. Trough serum concentration at steady state (Cminss) of Nivolumab 3. Cavgd28 of Relatlimab

7.2 Secondary Endpoints 1. Objective Response Rate (ORR) by Blinded Independent Central Review (BICR) per RECIST v1.1 2. Trough serum concentration at Day 28 after the first dose (Cmind28) of Nivolumab and Relatlimab 3. Peak serum concentration after the first dose (Cmax1) of Nivolumab and Relatlimab

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via HCP exams, patient reporting, and laboratory tests. **Serious Adverse Events (SAEs):** Standard oncology reporting timelines (typically within 24 hours of investigator awareness). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) overseen by the sponsor to routinely evaluate patient safety and exposure risks.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for objective efficacy endpoints; Safety Set for AE/SAE tabulations; Pharmacokinetic (PK) Population for drug exposure levels. **Sample Size Justification:** Target enrollment of $N=579$ is powered to statistically demonstrate non-inferiority in PK parameters between the subcutaneous and intravenous cohorts. **Handling of Missing Data:** Standard imputation approaches as designated by the protocol for missing tumor assessments or PK profiles.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring for protocol adherence and safety validations. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with source data

verification, blinded independent imaging read confirmations, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CA224-127 **Registry Number:** NCT05625399 **Posting ID:** 31774509539330535 **Sponsor/Lead Organization:** Bristol Myers Squibb
Study Title: A Study of Subcutaneous Nivolumab + Relatlimab Fixed-dose Combination (FDC) in Previously Untreated Metastatic or Unresectable Melanoma **Official Regulatory Title:** A Phase 3, Randomized, Open-label, Study of Subcutaneous Nivolumab + Relatlimab Fixed-dose Combination versus Intravenous Nivolumab + Relatlimab Fixed-dose Combination in Participants with Previously Untreated Metastatic or Unresectable Melanoma

12. Clinical Framework (Melanoma Specific)

Primary Condition: Melanoma (Disease Area: Melanoma; Malignant Melanoma; Preferred Name: melanoma; Preferred UMLS Name: Metastatic Melanoma) **Diagnosis Threshold:** Stage III (unresectable) or Stage IV (metastatic) **Condition Codes:** MeSH Code: D008545 | HPO Code: HP:0002861 | SNOMED CT: 372244006
Medical Methodology: Dual Immune Checkpoint Inhibition (Anti-PD-1 + Anti-LAG-3) **Optimization Target:** Pharmacokinetic and efficacy non-inferiority (SC vs IV)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced melanoma **Education Protocol (HCP):** Subcutaneous administration injection protocol and immune-related AE (irAE) management **Education Protocol (Patient):** Treatment schedule, adherence, and timely reporting of potential immune-related adverse events **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all RECIST v1.1 imaging readings

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Aligned with dosing cycles (e.g., Q4W) **Remote Monitoring Frequency:** Continuous irAE safety monitoring **Checklist Review Interval:** Tumor imaging assessments (CT/MRI) conducted at regular protocols intervals (e.g., every 8-12 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					